

From Building a RAG System to Understanding Its Limits: Reflections on Evidence Quality and Controlled Evaluation

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Abstract

This paper reflects on my learning in an NLP course project on retrieval-augmented generation (RAG) for science-domain multiple-choice question answering. The group project used SciQ to study how retrieval design choices and evidence quality affect a local, reproducible RAG pipeline. My individual contribution focused on moving the project from a minimal working system to a controlled evaluation framework with ablations, significance testing, multi-seed validation, and context interventions. The central lesson I draw from this work is that building a functioning RAG system is only the first step. I learned to treat the project as a study in which claims depend on comparison design, not just on whether a pipeline can be made to run. I also learned that evidence quality is a central bottleneck in RAG: a language model can benefit from retrieved support only when the decisive evidence is present in a usable form. Finally, I learned that interpreting results is harder than producing them, because sound conclusions depend on baselines, comparators, and precise descriptions of what the experiments actually manipulate. These lessons changed how I understand both RAG and empirical NLP work more broadly.

1 Introduction

Retrieval-augmented generation is commonly described as a practical way to supplement a language model with external evidence (Lewis et al., 2020; Gao et al., 2023). At first, I approached our project in exactly that spirit. The task seemed straightforward: build a local RAG system for science-domain multiple-choice question answering, compare several retrieval settings, and report which one worked best. However, the project became more educational when I realized that a RAG pipeline is not only a system to be built but also an experimental object whose behavior must be interpreted. Once I moved from the question “can this system answer

SciQ questions?” to the question “what explains the answers it gives?”, the project became a lesson in research design rather than only in implementation.

Our group used SciQ (Welbl et al., 2017) as a bounded evaluation setting and implemented a local-only RAG pipeline with dense retrieval over support paragraphs and a local generator constrained to output one option letter. The formal project question concerned retrieval design choices such as chunk size, top- k , and embedding model. My own learning, however, was shaped more by the evaluation process than by the architecture itself. The most important issues turned out to be whether the comparisons were controlled, whether evidence quality was being measured in a defensible way, and whether the conclusions in the paper matched what the code had actually tested. In that sense, the project helped me shift from a system-building mindset to an empirical one.

Rather than retelling the whole group report, I focus on the points where my own understanding changed. Three linked lessons emerged from that process. First, I learned to see a RAG project as a controlled study rather than merely an engineering task. Second, I learned that evidence quality was a stronger bottleneck than I had expected at the start. Third, I learned that a result only becomes interpretable when the comparison frame and the wording of the claim actually match the experiment.

2 The Project and My Role

The group project asked how retrieval design choices affect answer quality in a domain-specific RAG system. We used SciQ as the benchmark and treated the support paragraphs as the evidence source for retrieval. The baseline system combined sentence embeddings, vector search, and a local text-to-text generator. Although this sounds like a

standard implementation task, the project quickly evolved into a more methodological study. We moved from a minimal working pipeline to controlled comparison, from controlled comparison to stability testing, and from stability testing to evidence interventions that were designed to explain why a given setting worked.

My own role was concentrated on that methodological escalation. I helped establish the local end-to-end loop, but more importantly I extended it into a workflow that produced per-question outputs, comparable runs, and interpretable summaries. In practical terms, my work progressed from making the system run, to making runs comparable, to making claims defensible. This trajectory matters because it mirrors how my understanding changed. I was not simply finishing tasks one by one; each new comparison forced me back into the code and back into the claims we thought we were making.

3 Examination of My Learning

3.1 Controlled Evaluation as a Research Skill

My first major lesson was that a RAG project only becomes intellectually meaningful when it is turned into a controlled study. At the beginning, a working end-to-end pipeline felt like the main objective. Once the minimal loop was running, however, I realized that this only answered a weak question. It showed that retrieval, prompting, and generation could be connected, but it did not explain what retrieval contributed or why one setting outperformed another. A working pipeline answered whether the components could be connected; it did not yet show what explanatory role retrieval actually played. The ablation study changed that. Instead of asking only which configuration produced the highest accuracy, I had to ask what remained fixed across runs and what exactly was being varied.

The ablation results in Table 1 illustrate why this mattered. On the 200-example validation subset, the LLM-only baseline achieved 0.45 accuracy, while the best single-run RAG setting reached 0.84. The strongest configuration used chunk size 256, top- $k = 3$, and the all-MiniLM-L6-v2 embedding model. That result by itself was useful but incomplete. What was more valuable for my learning was the pattern across settings. Increasing k from 1 to 3 improved performance, but increasing it further to 5 did not help and instead increased both latency and prompt length. Chunk size had a relatively small effect compared with retrieval depth

Setting	Accuracy
LLM-only baseline (1000, 3 seeds)	0.4373 $\pm$ 0.0101
Best RAG (chunk 256, $k = 3$)	0.8140 $\pm$ 0.0046
Best single-run ablation (200, seed 0)	0.8400
Oracle gold support	0.8727
Remove answer sentence	0.4823

Table 1: The main results that shaped my interpretation of the project.

in the tested range. The compact trend in Figure 2 captures that point more clearly than a single best number can. In retrospect, this is also where my reading of the RAG literature became more critical. Recent work emphasizes that RAG quality depends on many interacting design choices and that systematic best-practice evaluation is still needed (Li et al., 2025; Gao et al., 2023). This project taught me that end-task gains are informative only when the comparator design makes clear what was changed and what remained fixed.

This part of the project connected strongly to Lab 4, but in a more specific way than the phrase “evaluation is important” would suggest. In that lab, comparing LoRA, head-tuning, and stronger or weaker baselines trained me to pay attention to fair comparison, efficiency-performance trade-offs, and the danger of presenting a best run without a meaningful reference point. Because I had already seen how misleading model comparisons can become when the baseline is weak or the cost profile is ignored, I was more inclined in the RAG project to insist on an LLM-only baseline, to rerun the best setting across three seeds, and to use paired significance testing rather than reporting one favorable number. When we reran the main comparison on 1000 validation examples, the gap between the LLM-only baseline and the best RAG setting remained large and stable. On the smaller paired comparison, McNemar’s test also showed a strong statistical difference. What I learned from this is that evaluation design is not an accessory to model building. It is one of the core research skills in NLP because it determines which claims deserve trust.

3.2 Evidence Quality as a Bottleneck in RAG

The second lesson, and the one that changed my understanding of RAG most, came from the evidence-quality experiments. After we had already finished the main interventions, I felt that the story was basically complete: the system worked, the best setting looked stable, and the remaining task was

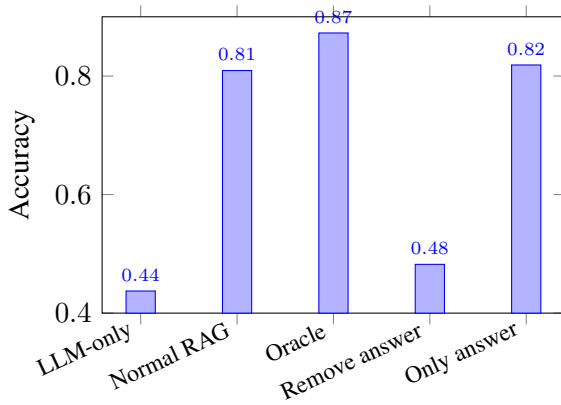


Figure 1: The main evidence-quality conditions. The answer-bearing sentence carries most of the useful signal.

mostly to write it up clearly. A turning point in my learning came when we added an oracle gold-support condition and lower-bound style evidence interventions. These additional controls changed how I interpreted our earlier results. Under the normal RAG setting, accuracy was 0.8093. When the oracle gold support paragraph was given directly to the generator, accuracy increased to 0.8727. When the answer-containing sentence was removed, accuracy dropped to 0.4823. By contrast, keeping only the answer sentence maintained near-baseline performance at 0.8187. These four conditions are shown in Figure 1. Before the oracle comparison, 0.8093 had looked close to “good enough” to me. Afterward, it looked like a middle point between a much clearer upper bound and a very informative lower bound.

This result changed my understanding in a very concrete way. The point of retrieval was not merely to supply additional text. The point was to make the decisive evidence available. When that evidence was removed, performance fell close to the LLM-only level. When only the answer-bearing sentence was preserved, performance stayed close to the normal RAG baseline. This suggested that the main source of improvement did not come from broad contextual richness, but from the availability of a small amount of highly relevant evidence. That interpretation is also consistent with the oracle condition, which improved over the normal RAG baseline but did not make the task trivial. Retrieval quality matters because it changes what the generator is allowed to condition on. At the same time, I would not generalize this too broadly: in our SciQ-style short-form setting, the answer-

bearing sentence carried most of the useful signal, but other tasks may require richer surrounding context. This caution is consistent with recent work showing that chunk granularity and context scope interact strongly with dataset characteristics (Bhat et al., 2025; Li et al., 2025).

Lab 2 helped me make sense of this pattern, but only after I saw the project results. In the lab, the causal mask and the greedy generation loop felt like fairly local implementation details: add the mask before softmax, take `model(context)[: , -1, :]`, append one token, and repeat. In the RAG project, those details came back in a much more concrete way. Once I saw that removing the answer sentence nearly erased the RAG gain, the lab code gave me a simple explanation: the model can only condition on the tokens that survive retrieval and actually enter the prompt. If the decisive sentence is missing, fluency does not bring it back. If that sentence is present, even without most of the surrounding paragraph, the model often has enough to choose the right option. That was the moment when Lab 2 stopped feeling like a separate transformer exercise and started to explain our oracle and intervention results. Lab 1 also became more meaningful in retrospect, because chunking and embeddings now looked less like isolated techniques and more like upstream decisions about whether the decisive sentence is retrievable at all.

3.3 Interpreting Results and the Limits of Conclusions

The third lesson was that a result does not speak for itself; it only becomes meaningful inside the right comparison frame. I first noticed this in the context-intervention suite. It would have been easy to summarize the distractor experiment by saying that irrelevant context hurts RAG. But the relevant comparison was between the top-1 baseline at 0.7750 and the top-1-plus-random-distractor setting at 0.7660. That is a small drop, not a collapse. The order-shuffling condition was even less dramatic: shuffling the top-3 contexts gave 0.8187, very close to the normal top-3 baseline of 0.8093. Those results did not support a broad story that the system was fragile. They supported a narrower one: in our setup, removing decisive evidence hurt a great deal, while adding one noisy paragraph or reordering the same evidence hurt much less.

The same issue appeared when I compared answer metrics with retrieval metrics. In the multi-seed validation setting, the best RAG configuration

achieved 0.8140 answer accuracy, while Recall@3 was 0.6170 and MRR was 0.5315. At first those numbers looked inconsistent to me. After checking the per-question outputs, they stopped looking inconsistent and started looking layered. There were cases where the exact gold support was missed but the model still answered correctly because a near-duplicate sentence carried the needed fact, as in the earthquake question where the retrieved context still said that earthquakes happen at plate boundaries even though the official gold paragraph was not retrieved. That made me more careful about what each metric could and could not prove. Retrieval metrics tell me whether the designated support was found. Answer accuracy tells me whether the generator could solve the question with the evidence it actually saw. They are related, but they are not interchangeable.

Writing became part of the same interpretive problem. While revising the paper, I noticed that I was describing some interventions as if we had corrupted the gold support paragraph itself. That was not what the code did. The code altered retrieved contexts, while the oracle experiment injected the gold support directly. Once I saw that mismatch, I had to rewrite the claim, not just polish the sentence. That was the point where writing stopped feeling like final cleanup. If I described the manipulation carelessly, I was also overstating what we had actually shown.

4 What I Learned, How I Learned It, and Why It Matters

What I learned. Looking back, the project taught me three tightly connected lessons. I learned that evidence quality is a stronger determinant of performance than the mere presence of additional context. I learned that comparator design is what turns a system result into an interpretable research claim. And I learned that comparisons, metrics, and wording have to line up before a result can be interpreted with confidence.

How I learned it. I learned these lessons through a sequence that kept forcing me to reopen work I thought was already finished: first implementing the minimal loop, then turning it into controlled ablations, then rerunning the strongest setting across seeds, then testing intervention conditions, and finally revising the paper when the oracle upper bound and the retrieved-context interventions had to be described more carefully. More than once I

thought the hard part was over, only to find that a new comparison changed the meaning of an earlier result. That is why I stopped treating implementation, evaluation, and writing as separate stages. In this project, whenever one of them changed, the other two had to be checked again.

Why it matters. This matters to me beyond this particular course project. Before working on this paper, I would have been tempted to treat a strong final accuracy score as the main evidence that the project had succeeded. After this project, that standard feels incomplete. In future NLP work, I will ask earlier whether the comparator is fair, whether the variables have been isolated cleanly, and whether the wording of the final claim exceeds the scope of the experiment. I also think the lesson matters more broadly for RAG practice. In our SciQ setting, most of the gain came from getting the answer-bearing sentence into the prompt, not from surrounding the model with more text. That suggests that, at least for short-form QA, retrieval precision and evidence selection may matter more than simply increasing context volume. It also suggests that evaluation should separate retrieval quality from generation quality; otherwise a system can look strong without showing whether it found the right evidence, relied on a paraphrase, or guessed correctly from partial cues.

5 Conclusion

This project began with me trying to make a local RAG system work and ended with me caring more about what each comparison allowed us to claim. The most durable lesson I take from it is methodological: RAG performance only becomes meaningful when evidence quality, comparator design, and result wording are checked together. That lesson will shape how I approach future NLP work, and I think it also matters for RAG research more broadly: if the gain comes mainly from retrieving the decisive sentence, then system design and evaluation should make that mechanism visible.

References

- Sinchana Ramakanth Bhat, Max Rudat, Jannis Spiekermann, and Nicolas Flores-Herr. 2025. [Rethinking chunk size for long-document retrieval: A multi-dataset analysis](#). *arXiv preprint arXiv:2505.21700*.
- Yunfan Gao, Yueqing Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yao Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2023. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Siran Li, Linus Stenzel, Carsten Eickhoff, and Seyed Ali Bahrainian. 2025. [Enhancing retrieval-augmented generation: A study of best practices](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 6705–6717. Association for Computational Linguistics.
- Johannes Welbl, Nelson F. Liu, and Matt Gardner. 2017. [Crowdsourcing multiple choice science questions](#). In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, pages 94–106. Association for Computational Linguistics.

A Supplementary Results

Setting	Accuracy	Δ vs. baseline
LLM-only	0.4373	-0.3720
RAG baseline	0.8093	—
Oracle gold support	0.8727	0.0634
Remove answer sentence	0.4823	-0.3270
Only answer sentence	0.8187	0.0094
Add distractor context	0.7660	-0.0433
Shuffle context order	0.8187	0.0094

Table 2: Results under different settings

Statistic	Value
Baseline accuracy	0.4500
Best RAG accuracy	0.8400
Accuracy delta	0.3900
95% CI for delta	[0.3150, 0.4650]
McNemar exact p -value	2.68971e-19

Table 4: Paired significance test for the 200-example baseline-versus-best comparison.

Embedding model	Chunk size	Top- k	Accuracy	Recall@ k	E2E latency (ms)
LLM-only	—	—	0.450	—	34.7
all-MiniLM-L6-v2	256	1	0.765	0.485	55.3
all-MiniLM-L6-v2	256	3	0.840	0.610	91.9
all-MiniLM-L6-v2	256	5	0.830	0.680	148.9
all-MiniLM-L6-v2	512	1	0.760	0.485	48.9
all-MiniLM-L6-v2	512	3	0.820	0.615	93.6
all-MiniLM-L6-v2	512	5	0.805	0.680	152.0
all-MiniLM-L6-v2	1024	1	0.760	0.485	48.3
all-MiniLM-L6-v2	1024	3	0.820	0.615	93.6
all-MiniLM-L6-v2	1024	5	0.805	0.680	153.8
msmarco-MiniLM-L6-v3	256	1	0.790	0.535	60.1
msmarco-MiniLM-L6-v3	256	3	0.830	0.660	91.7
msmarco-MiniLM-L6-v3	256	5	0.830	0.675	160.0
msmarco-MiniLM-L6-v3	512	1	0.785	0.525	48.5
msmarco-MiniLM-L6-v3	512	3	0.815	0.655	90.7
msmarco-MiniLM-L6-v3	512	5	0.795	0.670	161.6
msmarco-MiniLM-L6-v3	1024	1	0.785	0.525	52.4
msmarco-MiniLM-L6-v3	1024	3	0.815	0.655	92.0
msmarco-MiniLM-L6-v3	1024	5	0.795	0.670	160.3

Table 3: Results of the retrieval ablation study

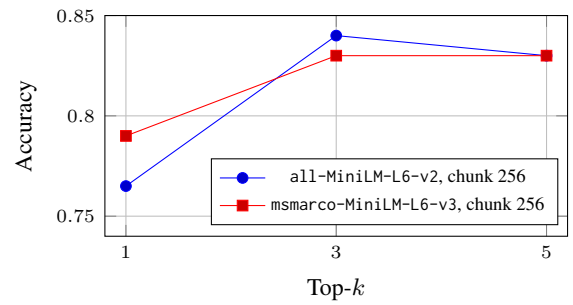


Figure 2: Top- k trend for the chunk-256 ablation. The strongest gain comes from moving from $k = 1$ to $k = 3$.